

Transformer Architecture: Quick Revision

Brief explanation + masked self-attention and cross-attention numerical

Deep Learning and GenAI TA Session

1. What is a Transformer?

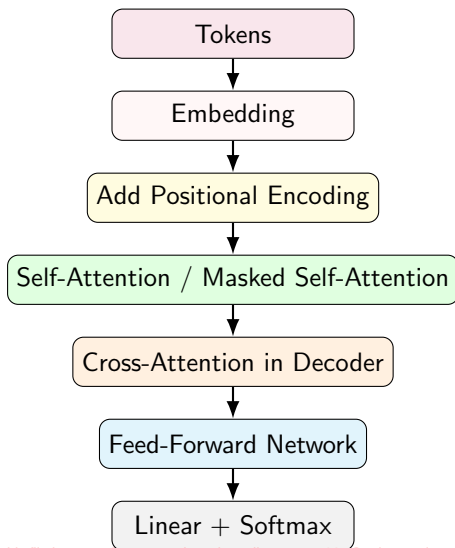
- A **Transformer** is a neural network architecture mainly used for sequence data such as text.
- Earlier models like RNNs and LSTMs read tokens one by one:

$$x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow \dots$$

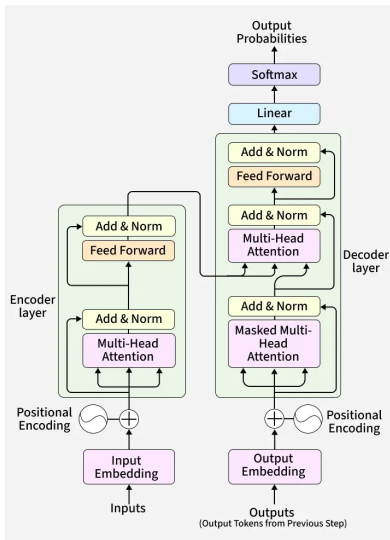
- A Transformer processes tokens more in parallel using **self-attention**.
- Main idea: each word looks at other words and decides which words are important.

Example: In “The cat sat on the mat”, the word **cat** may attend strongly to **sat**, because it helps understand what the cat is doing.

2. Step-by-Step Transformer Flow



4. Transformer Architecture Diagram



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5. Input Embedding and Positional Encoding

- A model cannot directly process words as text.
- Each token is converted into a numerical vector called an **embedding**.

$$x_i = \text{Embedding}(w_i)$$

Example:

$$\text{"cat"} \rightarrow [0.2, -0.1, 0.7, \dots]$$

Self-attention alone does not know word order, so we add positional encoding:

$$z_i = x_i + p_i$$

where x_i is the token embedding and p_i is the positional encoding.

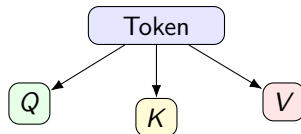
6. Self-Attention: Q , K , and V

- Self-attention allows each token to look at other tokens in the same sequence.
- Each token creates three vectors:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

Meaning

- Q : what I am looking for
- K : what information I contain
- V : actual information passed forward



7. Self-Attention Formula

The scaled dot-product attention formula is:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

For token i , the attention weight on token j is:

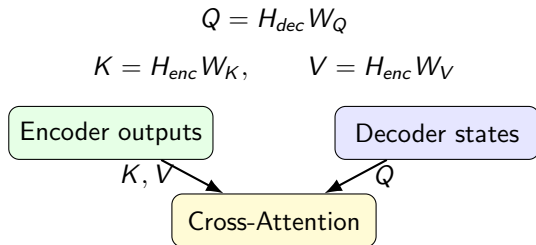
$$\alpha_{ij} = \frac{\exp \left(\frac{q_i k_j^T}{\sqrt{d_k}} \right)}{\sum_{r=1}^T \exp \left(\frac{q_i k_r^T}{\sqrt{d_k}} \right)}$$

Then the output representation for token i is:

$$z_i = \sum_{j=1}^T \alpha_{ij} V_j$$

8. Cross-Attention: Where Do Q , K , and V Come From?

- Cross-attention appears in the **decoder** of an encoder-decoder Transformer.
- It allows the decoder to attend to the encoder output.



Difference from self-attention

In self-attention, Q, K, V come from the same sequence. In cross-attention, Q comes from the decoder, while K, V come from the encoder.

9. Multi-Head Attention

- Instead of doing attention once, the Transformer does attention multiple times.
- Each attention unit is called a **head**.

For head h :

$$\text{head}_h = \text{Attention}(Q_h, K_h, V_h)$$

Then all heads are combined:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W_O$$

Why multiple heads?

One head may focus on grammar, another on meaning, and another on long-distance relations.

10. Add & Norm and Feed-Forward Network

After multi-head attention:

$$z = \text{LayerNorm}(x + \text{MHA}(x))$$

After the feed-forward layer:

$$y = \text{LayerNorm}(z + \text{FFN}(z))$$

The feed-forward network is:

$$\text{FFN}(x) = W_2 \phi(W_1 x + b_1) + b_2$$

Simple intuition

Attention mixes information across tokens. The feed-forward layer improves each token representation individually. Residual connections help gradients flow through deep networks.

11. Practice Question: Masked Self-Attention and Cross-Attention

In a decoder's masked self-attention layer, suppose the sequence length is $T = 4$ and the raw attention score matrix before masking is:

$$E = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 2 & 5 & 1 & 3 \\ 4 & 2 & 6 & 1 \\ 3 & 1 & 2 & 7 \end{bmatrix}$$

- (a) Write the masked score matrix $\tilde{E}$ that enforces causal autoregressive masking.
- (b) Compute the row-wise softmax of $\tilde{E}$ to obtain the attention weight matrix A . Treat $\exp(-\infty) = 0$.

12. Solution (a): Causal Masking

Causal masking means each token can only attend to itself and previous tokens.

$$\tilde{E} = \begin{bmatrix} 1 & -\infty & -\infty & -\infty \\ 2 & 5 & -\infty & -\infty \\ 4 & 2 & 6 & -\infty \\ 3 & 1 & 2 & 7 \end{bmatrix}$$

Meaning

Row i can only use columns $1, 2, \dots, i$. Future columns are blocked using $-\infty$.

13. Solution (b): Row-wise Softmax

Row 1:

$$\text{softmax}(1, -\infty, -\infty, -\infty) = (1, 0, 0, 0)$$

Row 2:

$$e^2 = 7.39, \quad e^5 = 148.41, \quad \text{sum} = 155.80$$

$$A_2 = \left(\frac{7.39}{155.80}, \frac{148.41}{155.80}, 0, 0 \right) \approx (0.047, 0.953, 0, 0)$$

Row 3:

$$e^4 = 54.60, \quad e^2 = 7.39, \quad e^6 = 403.43, \quad \text{sum} = 465.42$$

$$A_3 = \left(\frac{54.60}{465.42}, \frac{7.39}{465.42}, \frac{403.43}{465.42}, 0 \right) \approx (0.117, 0.016, 0.867, 0)$$

Row 4:

$$e^3 = 20.09, \quad e^1 = 2.72, \quad e^2 = 7.39, \quad e^7 = 1096.63, \quad \text{sum} = 1126.83$$

$$A_4 \approx (0.018, 0.002, 0.007, 0.973)$$

14. Solution (b): Final Attention Matrix

So the attention matrix is approximately:

$$A \approx \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.047 & 0.953 & 0 & 0 \\ 0.117 & 0.016 & 0.867 & 0 \\ 0.018 & 0.002 & 0.007 & 0.973 \end{bmatrix}$$

Interpretation

The fourth token can attend to all positions because it has no future token in this length-4 sequence.
The first token can attend only to itself.

16. Quick Summary

- Embedding converts tokens into vectors.
- Positional encoding adds word-order information.
- Self-attention lets tokens attend to other tokens in the same sequence.
- Masked self-attention prevents the decoder from seeing future tokens.
- Cross-attention lets the decoder attend to encoder outputs.
- Multi-head attention captures different relationships in parallel.
- Add & Norm and FFN make the network stable and expressive.